SAFETY DATA SHEET

AICAR — 50mg

SDS Version 1.0 | Date Prepared: 2026-04-15

Section 1 — Product and Company Identification

Product Name	AICAR — 50mg
Chemical Name	5-Aminoimidazole-4-carboxamide-1-beta-D-ribofuranoside
Synonyms	AICAR; Acadesine; AICA riboside
Catalog / SKU	aicar
Product Category	Growth & Recovery
Recommended Use	Laboratory research use only. Not for human or veterinary consumption.
Supplier	Helix Amino
Address	3199 Albany Post Rd, Suite 301G, Buchannan NY 10511
Website	https://helixamino.com
Email	info@helixamino.com
Emergency Phone	1-800-222-1222 (US Poison Control, 24 h)

Section 2 — Hazard Identification

GHS Classification: Classified under GHS Rev 3 as a mild skin and eye irritant. Category 3 (mild).

Signal Word: Warning

GHS Pictogram:

GHS07 — Exclamation mark (Warning)

Hazard Statements: H315 May cause mild skin irritation. H319 May cause mild eye irritation.

Precautionary Statements: P264 Wash thoroughly after handling. P280 Wear protective gloves, eye protection, and a lab coat. P305+P351+P338 If in eyes, rinse cautiously with water for several minutes. P302+P352 If on skin, wash with plenty of water.

Section 3 — Composition / Information on Ingredients

Component	CAS Number	% by Weight	Notes
5-Aminoimidazole-4-carboxamide-1-beta-D-ribofuranoside	2627-69-2	≥ 95%	Purity verified by HPLC

AMPK activator via conversion to ZMP. Cell-permeable adenosine analog.

Molecular Formula	C9H14N4O5
Molecular Weight	258.23 g/mol
Data Source	Confirmed against PubChem (NIH NLM)

Verification Note: *Molecular formula and weight confirmed against PubChem on 2026-04-15.*

Section 4 — First-Aid Measures

Inhalation: Remove to fresh air. Seek medical attention if symptoms develop or persist.

Skin Contact: Wash affected area thoroughly with soap and water. Remove contaminated clothing. Seek medical attention if irritation persists.

Eye Contact: Rinse immediately with plenty of water for at least 15 minutes, holding eyelids open. Seek medical attention.

Ingestion: Do not induce vomiting. Rinse mouth with water. Seek immediate medical advice. Contact Poison Control (1-800-222-1222).

Most Important Symptoms: No specific symptoms known. May cause mild irritation to eyes, skin, and respiratory tract.

Section 5 — Fire-Fighting Measures

Suitable Extinguishing Media: Water spray, dry chemical, carbon dioxide (CO_2), or alcohol-resistant foam.

Unsuitable Extinguishing Media: None known.

Specific Hazards: Combustion may produce carbon oxides, nitrogen oxides, and other toxic decomposition products.

Protective Equipment for Firefighters: Wear self-contained breathing apparatus and full protective clothing.

Section 6 — Accidental Release Measures

Personal Precautions: Avoid dust formation and inhalation. Wear appropriate personal protective equipment (gloves, goggles, lab coat, dust mask).

Environmental Precautions: Prevent release to drains, waterways, or the environment.

Containment and Cleanup: Carefully collect material and place in a suitable, labeled container for disposal. Clean affected area with water. Ventilate the area.

Section 7 — Handling and Storage

Handling: Use only in a well-ventilated laboratory environment. Avoid dust formation. Avoid contact with skin, eyes, and clothing. Do not eat, drink, or smoke when handling. Wash hands thoroughly after handling.

Storage: Store lyophilized at -20°C, protect from light and moisture. Reconstituted: store at 2-8°C, use within 30 days.

Incompatibilities: Strong oxidizing agents, strong acids, strong bases.

Section 8 — Exposure Controls / Personal Protection

Exposure Limits: No occupational exposure limits (OELs) have been established for this compound.

Engineering Controls: Use in a fume hood or equivalent local exhaust ventilation. Ensure adequate general ventilation.

Personal Protective Equipment (PPE):

- Respiratory protection: Dust mask or NIOSH-approved respirator if dust is generated.
- Hand protection: Chemical-resistant nitrile gloves.
- Eye protection: Chemical safety goggles or safety glasses with side shields.
- Skin and body protection: Lab coat and closed-toe footwear.

Section 9 — Physical and Chemical Properties

Appearance	White to off-white lyophilized powder
Odor	Odorless to faint characteristic odor
Odor Threshold	Not available
pH	Not determined
Melting / Freezing Point	Not determined
Initial Boiling Point and Boiling Range	Not applicable (solid)
Flash Point	Not applicable (non-flammable solid)
Evaporation Rate	Not applicable (solid)
Flammability (solid, gas)	Not flammable under normal conditions
Upper / Lower Flammability or Explosive Limits	Not available
Vapor Pressure	Not applicable (solid)
Vapor Density	Not applicable (solid)
Relative Density	Not determined
Solubility(ies)	Limited aqueous solubility; soluble in DMSO, ethanol

Partition Coefficient: n-octanol/water (log Kow)	Not available
Auto-ignition Temperature	Not available
Decomposition Temperature	Not available
Viscosity	Not applicable (solid)

Section 10 — Stability and Reactivity

Reactivity: No reactivity hazards expected under normal storage and handling conditions.

Chemical Stability: Stable under recommended storage conditions.

Possibility of Hazardous Reactions: None known under normal conditions.

Conditions to Avoid: Heat, direct sunlight, moisture, freeze/thaw cycles (for reconstituted material).

Incompatible Materials: Strong oxidizing agents, strong acids, and strong bases.

Hazardous Decomposition Products: Carbon oxides (CO, CO_2), nitrogen oxides (NO_x), and other toxic organic fragments on combustion.

Section 11 — Toxicological Information

Routes of Exposure: Inhalation of dust, skin contact, eye contact, accidental ingestion.

Acute Toxicity: No specific toxicological data available for this compound. Not tested for human or animal toxicity by the supplier. LD50 and LC50 values: Not available.

Skin Corrosion / Irritation: May cause mild irritation on prolonged contact.

Serious Eye Damage / Irritation: May cause mild irritation.

Respiratory or Skin Sensitization: No data available.

Germ Cell Mutagenicity: No data available.

Carcinogenicity: Not listed by IARC, NTP, or OSHA as a carcinogen.

Reproductive Toxicity: No data available.

STOT — Single Exposure: No data available. Not expected to cause specific target organ toxicity from a single exposure based on the available information.

STOT — Repeated Exposure: No data available. Not expected to cause specific target organ toxicity from repeated exposure based on the available information.

Aspiration Hazard: Not applicable. Product is supplied as a solid or lyophilized powder; aspiration classification criteria apply to liquid products only.

NOTE: This product is supplied for laboratory research use only. Toxicological data is limited. Handle as a potentially hazardous material.

Section 12 — Ecological Information

Ecotoxicity: No ecotoxicological data available.

Persistence and Degradability: No data available.

Bioaccumulative Potential: No data available.

Mobility in Soil: No data available.

Other Adverse Effects: Avoid release to the environment.

Section 13 — Disposal Considerations

Waste Treatment: Dispose of contents and container in accordance with local, state/provincial, and federal regulations. Do not release to drains, waterways, or the environment. Contact a licensed hazardous waste disposal service if necessary.

Section 14 — Transport Information

UN Number	Not regulated
UN Proper Shipping Name	Not regulated for transport
Transport Hazard Class (DOT / IATA / IMDG)	Not classified
Packing Group	Not applicable
Environmental Hazards	Not a marine pollutant
Special Precautions	Keep dry. Protect from physical damage and temperature extremes.

Section 15 — Regulatory Information

- United States (OSHA): Complies with the OSHA Hazard Communication Standard (29 CFR 1910.1200).
- TSCA Inventory: Research use only. Not evaluated for TSCA listing.
- SARA Title III (Sections 302, 304, 311, 312, 313): Not listed.
- California Proposition 65: Not listed.
- Canada (WHMIS): Not classified as a controlled product.
- This product is sold for laboratory research use only and is not intended for human consumption, veterinary use, food use, or cosmetic use.

Section 16 — Other Information

SDS Version	1.0
Date Prepared	2026-04-15
Date of Last Revision	2026-04-15
Prepared By	Helix Amino Compliance Team
Issued By	Helix Amino, 3199 Albany Post Rd, Suite 301G, Buchannan NY 10511
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Reference Standards	29 CFR 1910.1200 Appendix D; ANSI Z400.1
Revision Summary	Initial issue.

Key to abbreviations: *CAS = Chemical Abstracts Service; GHS = Globally Harmonized System; HCS = Hazard Communication Standard; HPLC = High-Performance Liquid Chromatography; IARC = International Agency for Research on Cancer; LC50 = Lethal Concentration, 50%; LD50 = Lethal Dose, 50%; NIOSH = National Institute for Occupational Safety and Health; NTP = National Toxicology Program; OEL = Occupational Exposure Limit; OSHA = Occupational Safety and Health Administration; PPE = Personal Protective Equipment; STOT = Specific Target Organ Toxicity; SDS = Safety Data Sheet; TFA = Trifluoroacetic Acid; TSCA = Toxic Substances Control Act; WHMIS = Workplace Hazardous Materials Information System.*

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laboratory research purpose. This product is NOT intended for human consumption, veterinary use, diagnostic use, therapeutic use, or any in vivo use in humans or animals.

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